

IRIS CLASSIFICATION USING MACHINE LEARNING

Student Details:

Name: Matteddula Mayuri

College: Sri Venkateswara College of Engineering

Task Number: Task 1 – Iris Classification

Objective:

The objective of this project is to build a machine learning model that predicts the species of iris flowers using sepal and petal measurements.

Dataset:

The Iris dataset contains 150 samples and four features:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

The target variable is the iris species:

- Iris-setosa
- Iris-versicolor
- Iris-virginica

Exploratory Data Analysis:

Data analysis was performed using Pandas, Matplotlib, and Seaborn. The dataset contained no missing values. Scatter plots, histograms, and pair plots were used to visualize the relationships between features.

Models Used:

The following machine learning algorithms were implemented:

- Logistic Regression
- K-Nearest Neighbors (KNN)

- Decision Tree

The dataset was split into training and testing sets using an 80:20 ratio.

Results:

The models were evaluated using accuracy, precision, recall, and F1-score. KNN achieved the best performance and was selected as the final model. The trained model was saved as iris_model.pkl.

Conclusion:

This project successfully demonstrated the complete machine learning workflow including data analysis, visualization, model training, evaluation, and model saving.